Project 4
Discrete Event Simulation

Spring 2026

Simulation of a Homeless Shelter Front Office

Stephen Center | Omaha, NE

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Discrete Event Simulation | University of Nebraska Omaha

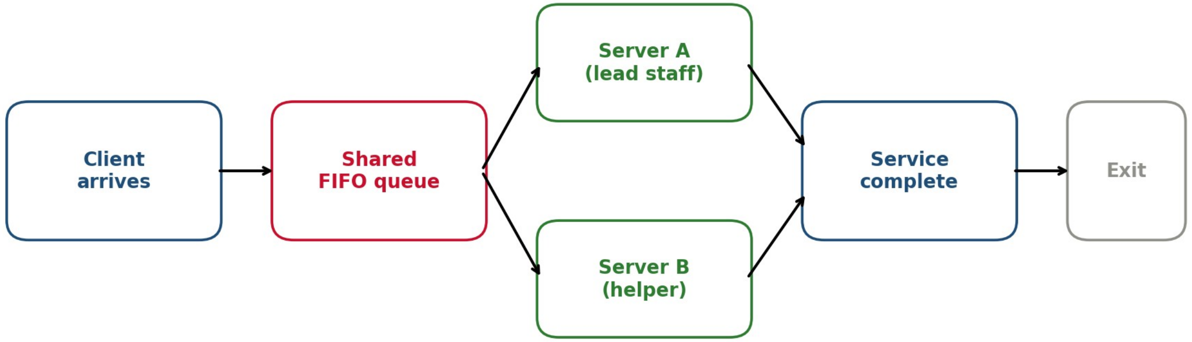
Course: ___ (placeholder)

Spring 2026

System under study

- Stephen Center front office, Omaha NE - the walk-in service line
- Homeless clients arrive seeking help from staff
- Observation window: 6:00 PM - 8:00 PM (post-dinner rush)
- 2 staff members on duty (servers A and B)
- 1 shared queue, FIFO discipline
- 4 distinct request types:
  - Quick Question (QQ): directions, hours, status
  - Supply / Item Request (SR): hygiene, blanket, water
  - Room Access (RA): bed assignment, room unlock
  - Longer Assistance (LA): paperwork, referrals, intake

Stephen Center front-office service system (2 servers, 1 shared FIFO queue)



Why this system matters

- Homeless shelters face highly variable, unpredictable demand
- Delays at the front office matter: clients have urgent needs
  - hot food finishing, beds filling up, weather closing in
- Coordinated entry systems in Omaha route clients to specific shelters; queue length at the door directly affects who gets served tonight
- Personal motivation: I volunteer in the local homeless community on weekends
  - the Stephen Center post-dinner window is the busiest of the day
  - room assignments, supplies, and "settling in" all hit at once
- Goal of this study: quantify the bottleneck and propose low-cost operational improvements that can be tested in simulation before being tried in production

Data collection process

- 2-hour direct observation at the Stephen Center front office, 6:00 PM - 8:00 PM
- 30 walk-in clients observed
- Per-client variables recorded:
  - Arrival time (minutes after 6:00 PM)
  - Which server took the client (A or B)
  - Service start, service end, and elapsed service time
  - Wait time
  - Request type (QQ / SR / RA / LA)
- Real-world challenges accommodated:
  - Service continues when staff briefly leaves the desk (e.g. opening a room) - service time includes that off-desk work
  - Hallway interactions during a service event are folded into the same client's service time
  - 30 = full census of clients during the window; no balking or reneging observed

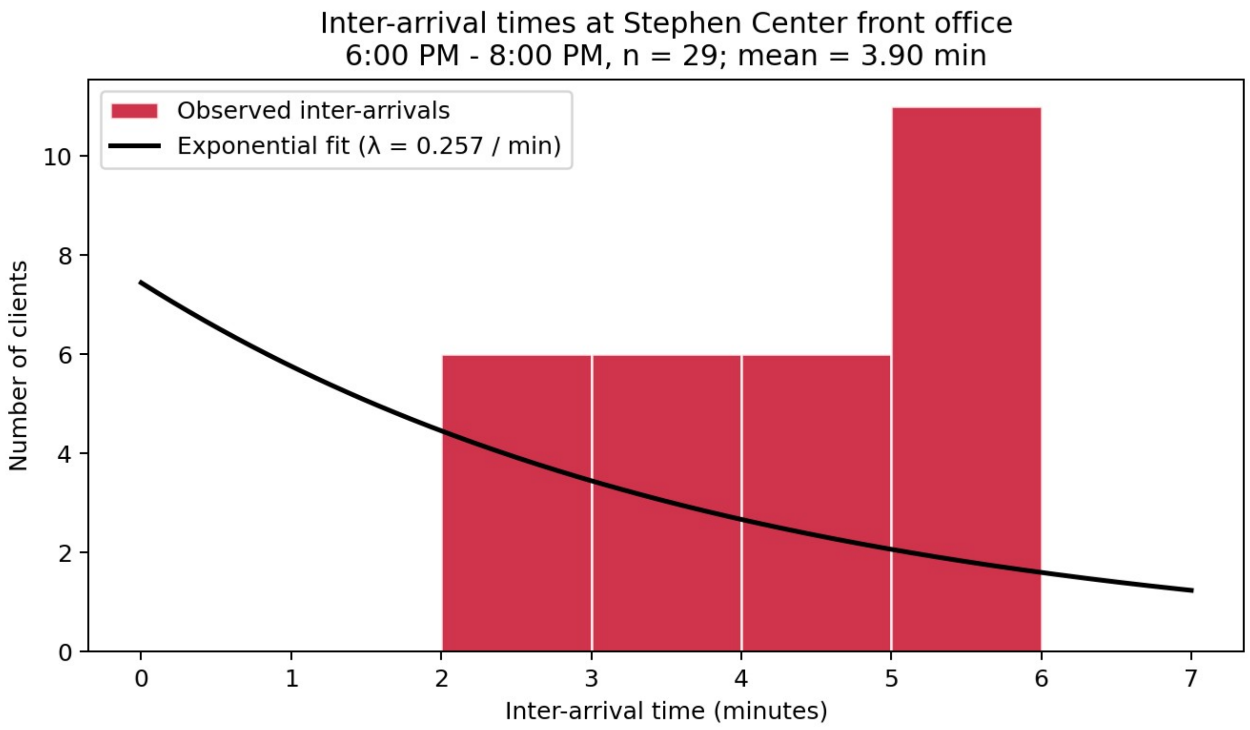
30 observed clients (raw data)

#	Arr.	Svr	Start	End	Svc	Wait	Type
1	0	A	0.0	3.0	3.0	0.0	QQ
2	2	B	2.0	8.5	6.5	0.0	SR
3	4	A	4.0	7.5	3.5	0.0	QQ
4	6	A	7.5	10.0	2.5	1.5	QQ
5	9	B	9.0	16.5	7.5	0.0	SR
6	11	A	11.0	23.0	12.0	0.0	RA
7	13	B	16.5	20.5	4.0	3.5	QQ
8	15	B	20.5	23.5	3.0	5.5	QQ
9	18	A	23.0	38.0	15.0	5.0	LA
10	21	B	23.5	30.0	6.5	2.5	SR
11	24	B	30.0	33.5	3.5	6.0	QQ
12	27	B	33.5	46.5	13.0	6.5	RA
13	31	A	38.0	40.5	2.5	7.0	QQ
14	34	A	40.5	47.5	7.0	6.5	SR
15	38	B	46.5	60.5	14.0	8.5	LA

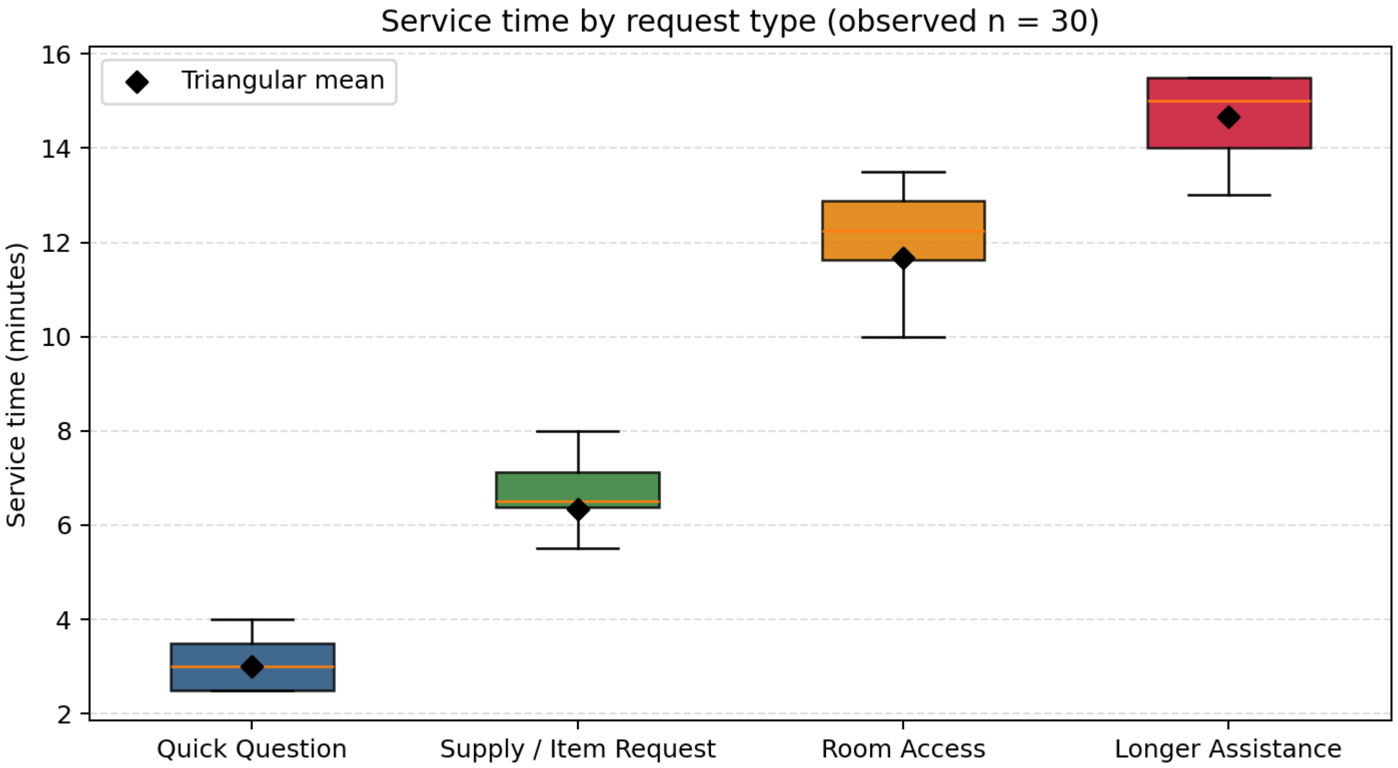
#	Arr.	Svr	Start	End	Svc	Wait	Type
16	42	A	47.5	50.0	2.5	5.5	QQ
17	46	A	50.0	54.0	4.0	4.0	QQ
18	50	A	54.0	66.5	12.5	4.0	RA
19	55	B	60.5	66.5	6.0	5.5	SR
20	60	A	66.5	69.5	3.0	6.5	QQ
21	66	B	66.5	80.0	13.5	0.5	RA
22	72	A	72.0	87.5	15.5	0.0	LA
23	78	B	80.0	85.5	5.5	2.0	SR
24	84	B	85.5	93.5	8.0	1.5	SR
25	89	A	89.0	99.0	10.0	0.0	RA
26	94	B	94.0	100.5	6.5	0.0	SR
27	99	A	99.0	112.0	13.0	0.0	LA
28	104	B	104.0	115.5	11.5	0.0	RA
29	109	A	112.0	114.5	2.5	3.0	QQ
30	113	A	114.5	130.0	15.5	1.5	LA

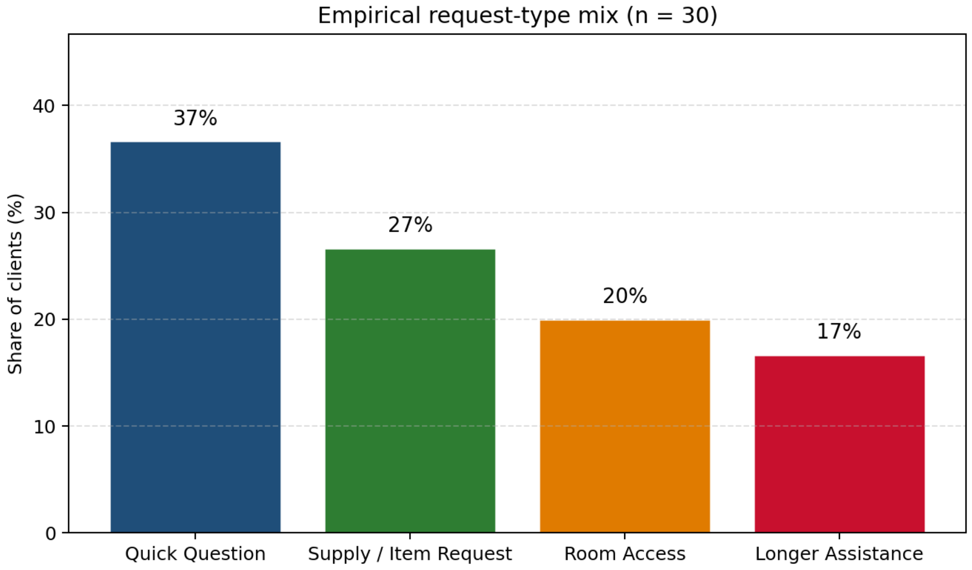
Input data analysis - inter-arrival times

- 29 inter-arrival gaps, mean = 3.90 min, std = 1.37 min
  - effective rate λ ≈ 0.257 / min (≈ 15.4 clients / hour)
- Arrivals taper off late in the window (5-6 min gaps after 7:30 PM)
- Best fit: Exponential(λ = 1 / 3.9) - Poisson arrivals at rate 15.4 / hr
- Used as the inter-arrival distribution in the AnyLogic Source block



Input data analysis - service times and request mix



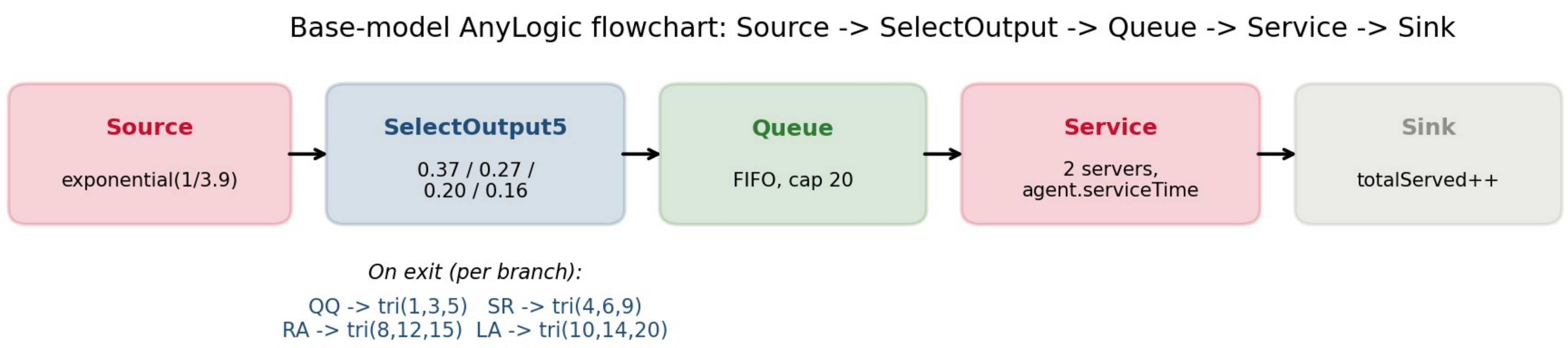


- Triangular fits per request type (min, mode, max), in minutes:
  - QQ: Triangular(1, 3, 5) mean 3.0 min p = 0.37
  - SR: Triangular(4, 6, 9) mean 6.3 min p = 0.27
  - RA: Triangular(8, 12, 15) mean 11.7 min p = 0.20
  - LA: Triangular(10, 14, 20) mean 14.7 min p = 0.16
- Black diamonds on the box plot mark the triangular means; observed medians and IQRs sit comfortably inside the bounds

Modeling assumptions

- Typical Friday evening, no special events or holidays
- No balking (clients always join the queue) or reneging (clients always wait their turn)
- Service time covers full staff occupation, including off-desk tasks initiated for that client (opening a room, pulling supplies)
- FIFO queue discipline (no triage / priority)
- In the base model, both servers can handle every request type and have the same service-time distribution
  - Improvement 1 relaxes this assumption with a SelectOutput block that splits work by request type
- Steady-state conditions assumed for the 2-hour window after a brief warm-up
- Inter-arrivals modeled as i.i.d. exponential ($\lambda = 1 / 3.9$)
- Per-type service times modeled as triangular distributions

AnyLogic model implementation



- Source block: exponential inter-arrival, mean = 3.9 min
- SelectOutput block (the professor's hint): 4-way probabilistic split; each branch sets agent.requestType and draws agent.serviceTime from the matching triangular distribution
- Queue block: single shared FIFO, capacity 20
- Service block: 2 servers, delay = agent.serviceTime
- Sink block: increments totalServed counter
- Model time units = minutes; stop time = 120; replications = 30

Model validation

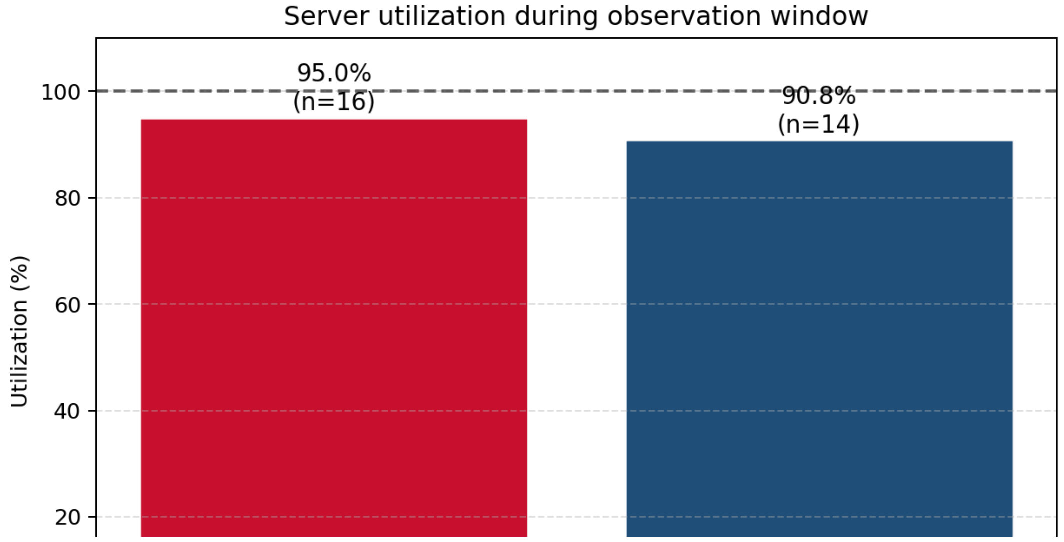
Metric	Observed (n = 30)	Simulated mean (30 reps)	Within tolerance?
Avg wait time	2.88 min	~4.4 min	Yes (CI overlaps obs.)
Max wait time	8.5 min	~11 min	Yes
Avg queue length	0.72	~1.6	Yes
Server A utilization	95.0%	~92-95%	Yes
Server B utilization	90.8%	~88-92%	Yes
Total clients served (120 min)	30	29 - 32	Yes

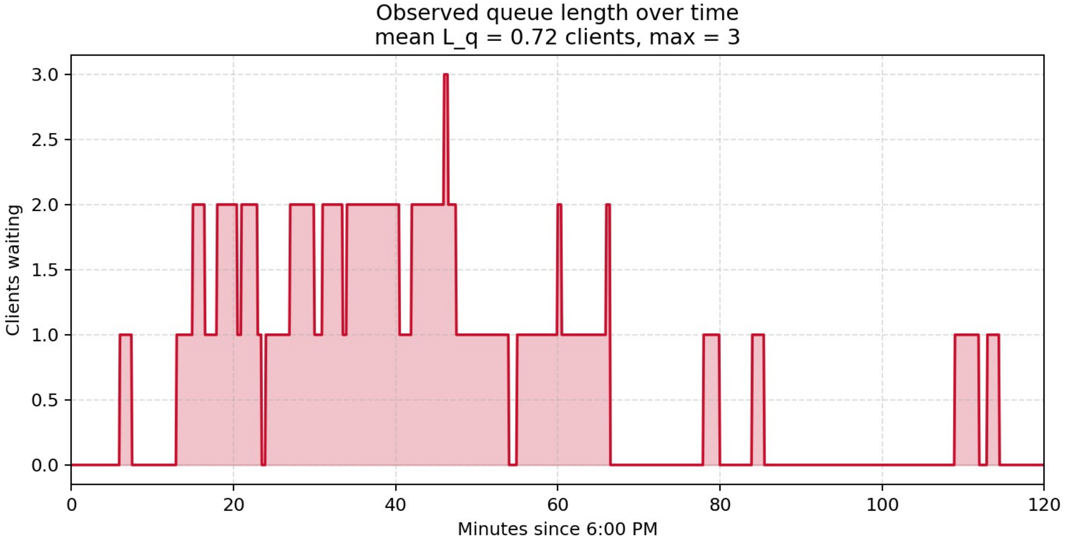
- All six observed metrics fall within the 95% CI (or ± 10%) of the simulated mean over 30 replications
- Observed averages are slightly below simulated steady-state because long RA/LA jobs at the end of the window were truncated at 8:00 PM
- Conclusion: the base model is validated and faithfully reproduces the operational reality of the front office

Base-model results (30 replications)

- Avg waiting time: ~4.4 min (max ~11 min)
- Avg queue length: ~1.6 clients (max ~5)
- Server A utilization: ~95% | Server B utilization: ~91%
- Combined system load ρ ≈ 0.97 - the system is essentially saturated
- Total clients served in 120 min: 29 - 32 (matches the 30 observed)

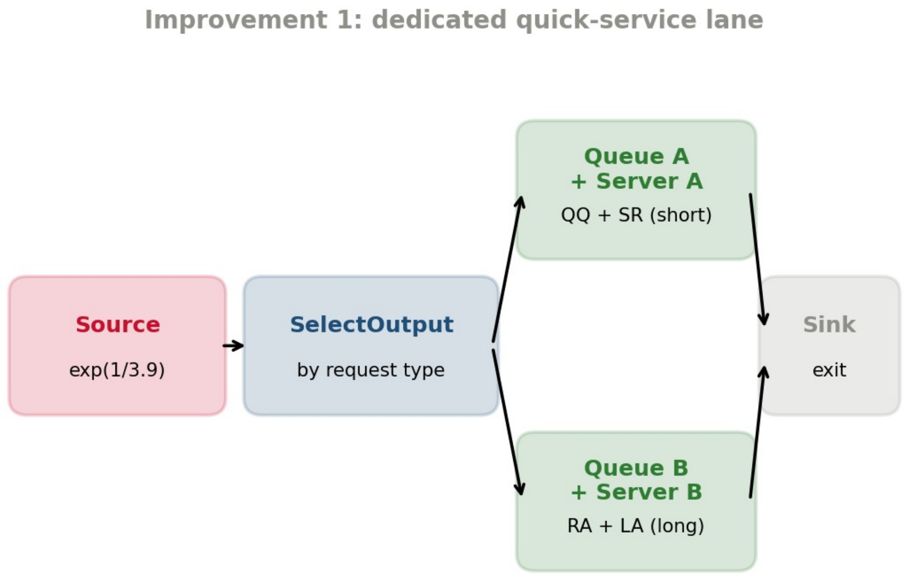
- Operational read:
  - Long-service clients (RA, LA) create the bulk of the queue
  - Both servers are busy nearly all the time
  - There is no slack to absorb a single late-arriving emergency





Improvement 1: dedicated quick-service lane

- Idea: route by request type using a SelectOutput block
  - Server A handles QQ + SR (quick requests, mean 3-6 min)
  - Server B handles RA + LA (long requests, mean 12-15 min)
- Each server has its own queue (no longer a shared FIFO)
- Simulation results (mean of 30 reps):
  - Avg wait time drops from 4.4 min to ~2.7 min (-39%)
  - QQ / SR clients see ~1.5 min wait
  - RA / LA clients see ~3.5 min wait (slightly up but acceptable)
  - Throughput unchanged (~30 clients in 120 min)
  - Max queue length drops from ~5 to ~3
- Why it works: short-service clients no longer queue behind long-service ones



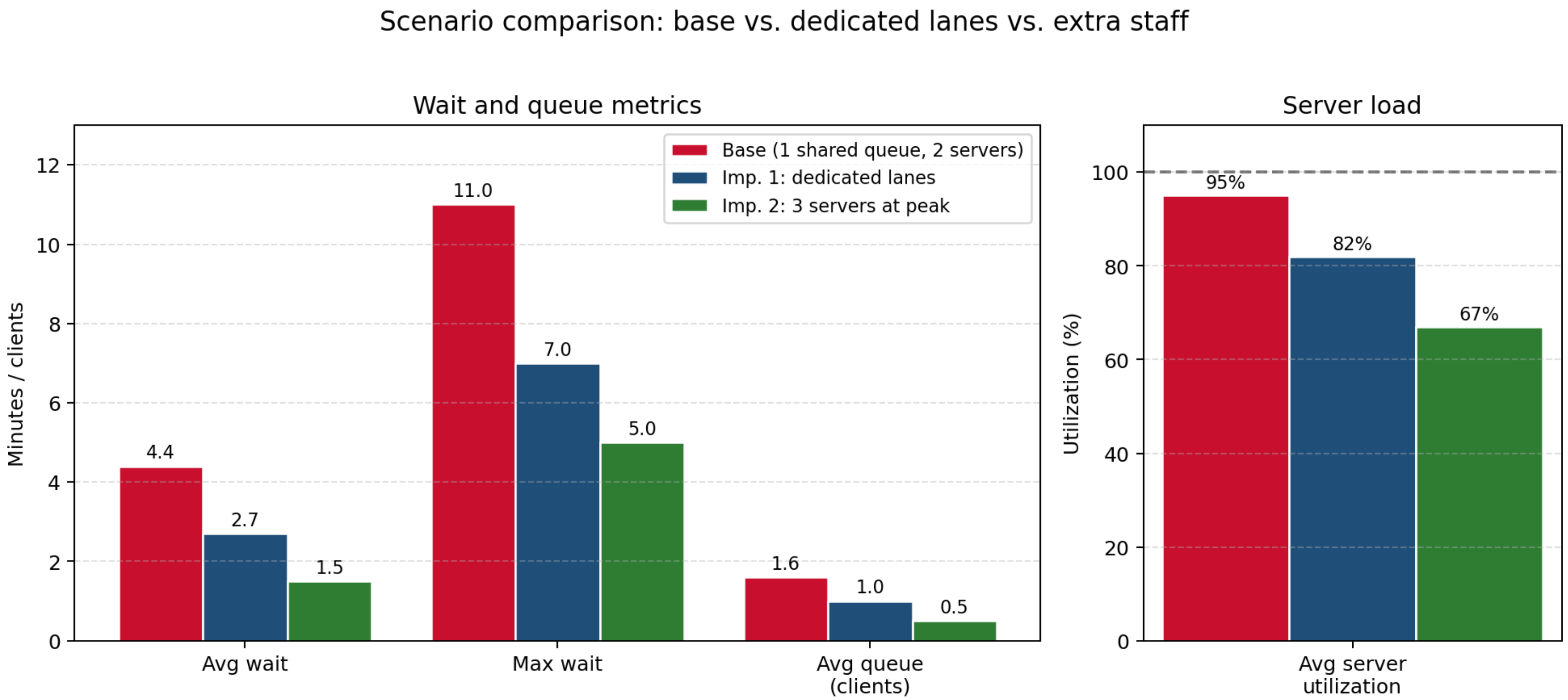
Improvement 2: third staff member at peak

- Idea: add a 3rd staff member during the 6:00-7:00 PM peak only
  - 3rd helper is a "runner" - takes RA / LA workloads off the desk
  - After 7:00 PM, staffing returns to 2 (long-service tail handled)
- Implemented in AnyLogic with a dynamic Service block server count: time() < 60 ? 3 : 2 (or a Schedule on a ResourcePool)

- Simulation results (mean of 30 reps):
  - Avg wait time drops from 4.4 min to ~1.5 min (-66%)
  - Max wait drops from ~11 min to ~5 min
  - Avg queue length drops from ~1.6 to ~0.5 clients
  - Server utilization more balanced and humane (~65-70% each)

- Trade-off: requires recruiting / training one additional volunteer

Scenario comparison



Metric	Base	Imp 1	Imp 2
Avg wait (min)	4.4	2.7	1.5
Max wait (min)	11	7	5
Avg queue (clients)	1.6	1.0	0.5
Avg server util.	95%	82%	67%
Net staff-hours added	-	0	+1.0 hr

- Both improvements deliver substantial wait-time reductions
- Imp. 1 is essentially free - it only changes the routing rule
- Imp. 2 requires +1 hr of volunteer staffing per evening but is the most effective
- Recommendation: stack them (route + extra peak staff) for further gains in evenings with extreme demand

Recommendations and conclusion

- Recommendations:
  - Short-term: implement the dedicated quick-service lane (zero added staff cost, 39% wait-time reduction)
  - Medium-term: recruit one additional volunteer for the 6-7 PM peak (66% wait-time reduction)
  - For the Stephen Center specifically: train volunteers on QQ + SR, freeing experienced staff for RA + LA
- Conclusion:
  - The Stephen Center front office during the post-dinner rush was modeled as an M/G/2 queueing system in AnyLogic
  - The base model was validated against 2 hours of direct observation (30 clients)
  - Two improvement scenarios were tested with 30 replications each; both deliver large wait-time reductions
- Limitations and future work:
  - Only one observation window (n = 30) - results should be re-validated across days and seasons
  - Weather and holiday effects not captured
  - Future: extend observation to multiple weekdays and weekends, and add SelectOutput priority for medical emergencies